

A quantity of 80.0 g of a salt (XY) of strong acid and weak base is dissolved in water to form 2.0 L aqueous solution. At 298 K, the pH of the solution is found to be 5.0. If XY forms CsCl type crystal and the radius of X^+ and Y^- ions are 160 pm and 186.4 pm, respectively, the the correct information is
(Given : K_b of $XOH = 4 \times 10^{-5}$, $N_A = 6 \times 10^{23}$, $\sqrt{3} = 1.732$)

✓ (A) molar mass of the salt is 100 g/mol.

✓ (B) the degree of hydrolysis of salt is 2.5×10^{-5} .

✓ (C) unit cell edge length of XY crystal is 400 pm.

✓ (D) density of solid XY is 2.6 g/cm³.

$$\begin{array}{r} 186.4 \\ 160.0 \\ \hline 346.4 \end{array}$$

$$d = 1 \times \frac{100}{6 \times 10^{23} \times (4 \times 10^{-8})^3} \frac{g}{cc} = \frac{100}{0.6 \times 64}$$

$$= \frac{100}{38.4} \approx 2.6 g/cc$$

Solⁿ: X^+Y^- | $X^+ \rightarrow$ hydrolysis | CsCl type crystal | $C = \frac{8}{100 \times 2} = 0.4 M$
80 g | 2 L
pH = 5 $K_b = 4 \times 10^{-5} \Rightarrow PK_b = 5 - \log 4$ | $r_{X^+} = 160$
 $r_{Y^-} = 186.4$ } pm.

Solⁿ: $pH = \frac{1}{2} (pK_w - pK_b - \log C)$
 $5 = \frac{1}{2} (14 - 5 + \log 4 - \log \frac{80}{M})$
 $10 = 9 + \log \left(\frac{4}{80} \times 2M \right)$
 $\log 10 = 1 = \log \frac{M}{10} \Rightarrow \frac{M}{10} = 10 \Rightarrow M = 100 g/mol.$

$h = \sqrt{\frac{K_w}{K_b \times C}} = \sqrt{\frac{10^{-14}}{4 \times 10^{-5} \times 0.4}} = \frac{1}{4} \times 10^{-4} = 2.5 \times 10^{-5}$
 $2r_+ + 2r_- = a\sqrt{3}$
 $a = \frac{2(160 + 186.4)}{1.732} = \frac{2 \times 346.4}{1.732} = 400 pm$

The crystalline structure of a solid is CCP for 'X' atoms. If 'Y' atoms occupy all the octahedral voids and 'Z' atoms occupy all the tetrahedral voids, without any distortion in the CCP arrangement of 'X' atoms, then the fraction of body diagonal not covered by the atoms is

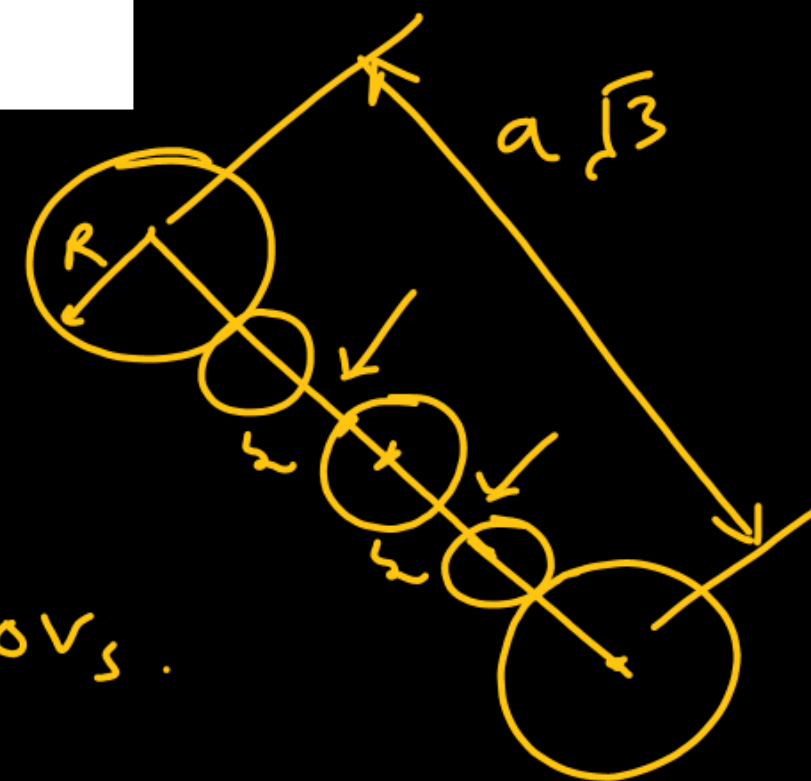
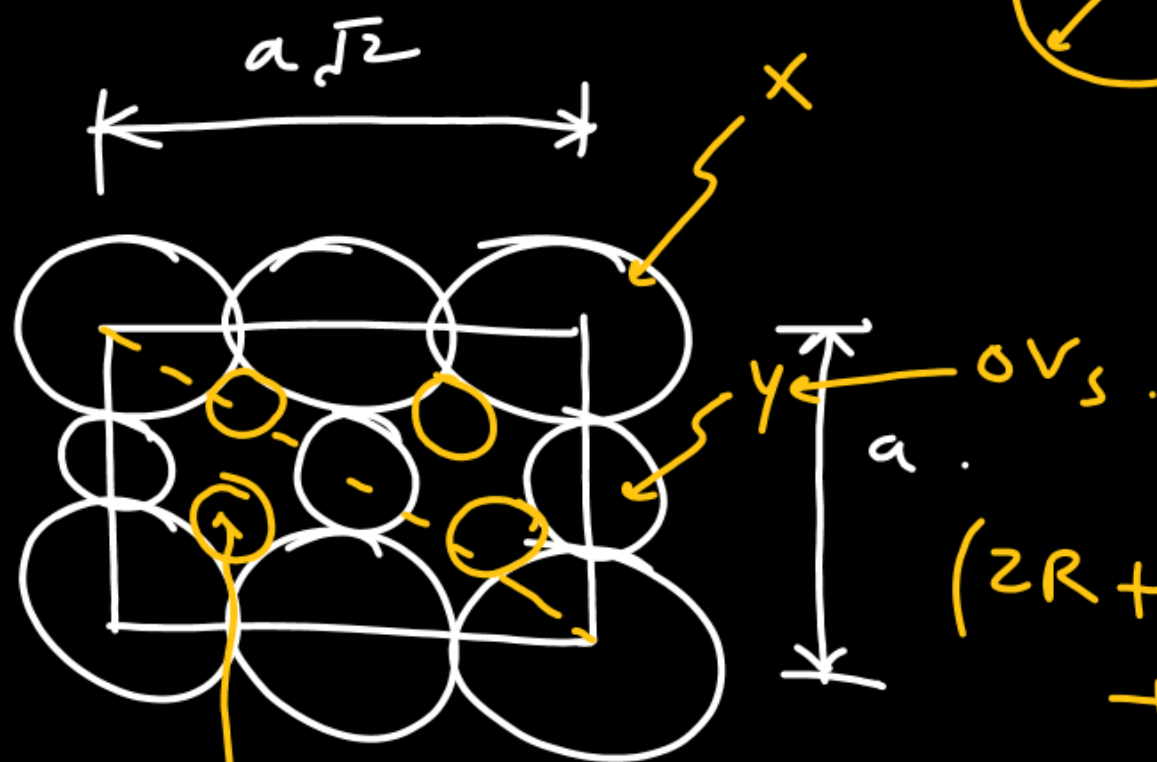
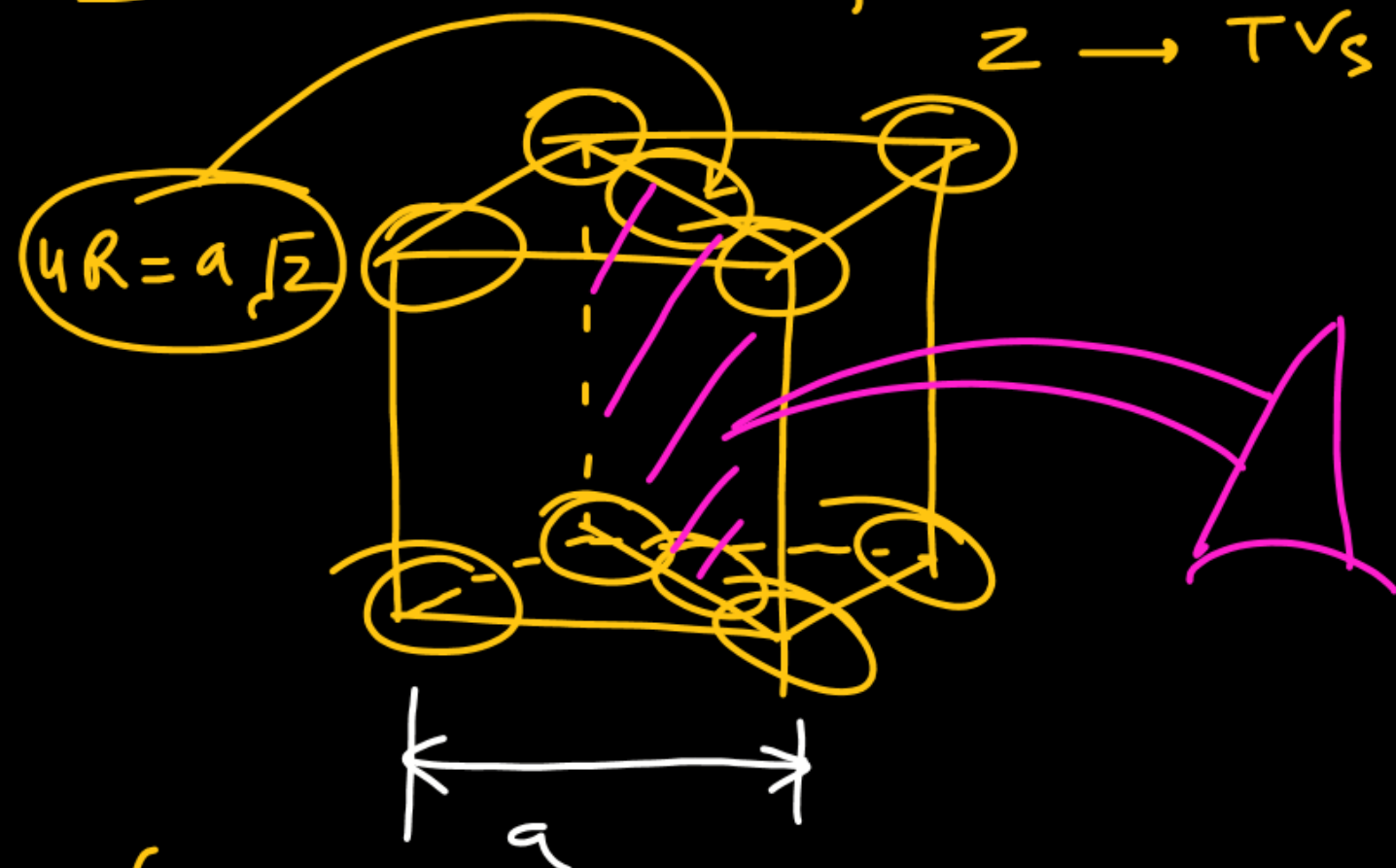
(A) 0.26

(B) 0.24

(C) 0.76

(D) 0.81

Solⁿ: X → CCP, Y → O.V.s
Z → T.V.s

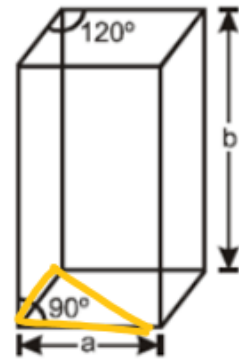


$$(2R + 4 \times 0.215R + 2 \times 0.414R + x) = a\sqrt{3}$$

fraction of body diagonal not covered = $\frac{Z(T.V.s) R \left(\frac{4 \times \sqrt{3}}{\sqrt{2}} - 3.72 \right)}{\sqrt{3} \times 4R / \sqrt{2}} \approx 0.24$

O.V. ⇒ 0.414R
T.V. ⇒ 0.215R

A_2B molecules (molar mass = 259.8 g/mol) crystallises in a hexagonal lattice as shown in figure. The lattice constants were $a = 5 \text{ \AA}$ and $b = 8 \text{ \AA}$. If density of crystal is 5 g/cm^3 then how many molecules are contained in given unit cell? (use $N_A = 6 \times 10^{23}$)



(A) 6

(B) 4

(C) 3

(D) 2

Solⁿ:

$$d = z \times \frac{M}{V_{\text{unit cell}}}$$

$$5 = z \times \frac{259.8}{6 \times 10^{23} \times \left(2 \times \frac{\sqrt{3}}{4} \times a^2\right) \times b}$$

$$5 = \textcircled{2} \times \frac{259.8}{0.3 \times 10^{25} \times \sqrt{3} \times (5 \times 10^{-8})^2 \times (8 \times 10^{-8})}$$

$$5 = z \times \frac{259.8}{0.3 \times 1.732 \times 25 \times 8}$$

$$z = \frac{5 \times 2.4 \times 1.732 \times 25}{259.8}$$

$$\Rightarrow z = 2$$

An ionic solid is HCP of Q^{2-} ions and P^{x+} ions are in half of the tetrahedral voids. The value of x should be

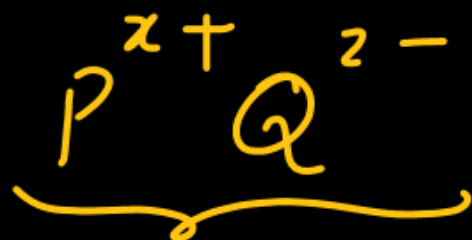
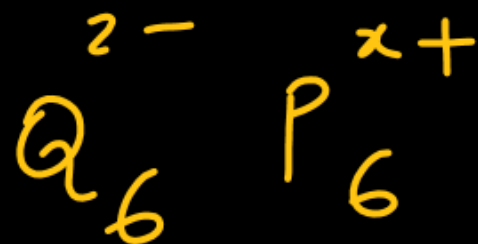
(A) 1

☒ (B) 2

(C) 4

(D) $1/2$

Solⁿ:



$$x = +2$$

A metal crystallizes in such a lattice in which only 70% of the total space of the crystal is occupied by the atoms. If the atomic mass of the metal is 32π g/mol and the atomic radius is 0.2 nm, the density of the metal is

(A) 7.0 g/cm^3

☒ (B) 3.5 g/cm^3

(C) 10.5 g/cm^3

(D) 14.0 g/cm^3

Solⁿ: P. efficiency = 70%.

$M = 32\pi \text{ g/mol}$, $r = 0.2 \times 10^{-9} \text{ m}$
 $= 2 \times 10^{-8} \text{ cm}$

density of metal = $\frac{\text{Molar mass}}{\text{volume of unit cells having 1 mol of element.}}$

$$= \frac{32\pi}{\frac{\left(\frac{4}{3}\pi r^3 \times NA\right)}{0.7}}$$

$$= \frac{32\pi \times 0.7}{\frac{4}{3}\pi \times (2 \times 10^{-8})^3 \times 6 \times 10^{23}}$$

$$= \frac{8 \times 3 \times 0.7}{8 \times 6 \times 0.1} = 7/2 = 3.5 \text{ g/cm}^3$$

Ans.

There are three cubic unit cells A, B and C. A is FCC and all of its tetrahedral voids are also occupied. B is also FCC and all of its octahedral voids are also occupied. C is simple cubic and all of its cubic voids are also occupied. If voids in all unit cells are occupied by the spheres exactly at their limiting radius, then order of packing efficiency would be

(A) $A < B < C$

☒ (B) $C < A < B$

(C) $C < B < A$

(D) $A < C < B$

Solⁿ: (A) In FCC,
 $4R = a\sqrt{2}$

FCC + All TVs
 (R) $\rightarrow 0.225R$

$$P.f = \frac{4 \times \frac{4}{3} \pi R^3 + 8 \times \frac{4}{3} \pi (0.225R)^3}{(4R/\sqrt{2})^3}$$

$C < A < B$

(B) $\rightarrow 0.414R$
 FCC + all OV_s

$$P.f = \frac{4 \times \frac{4}{3} \pi R^3 + 4 \times \frac{4}{3} \pi (0.414R)^3}{(4R/\sqrt{2})^3}$$

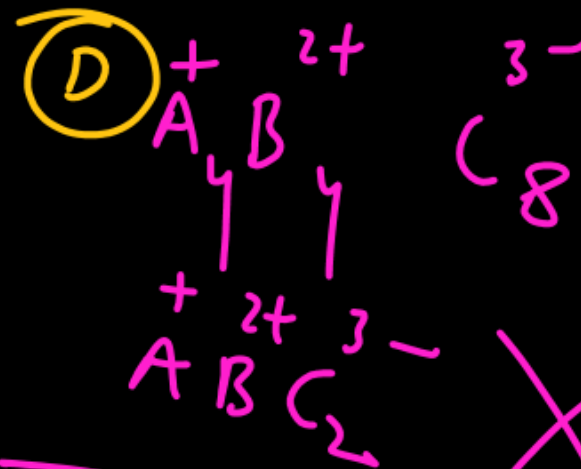
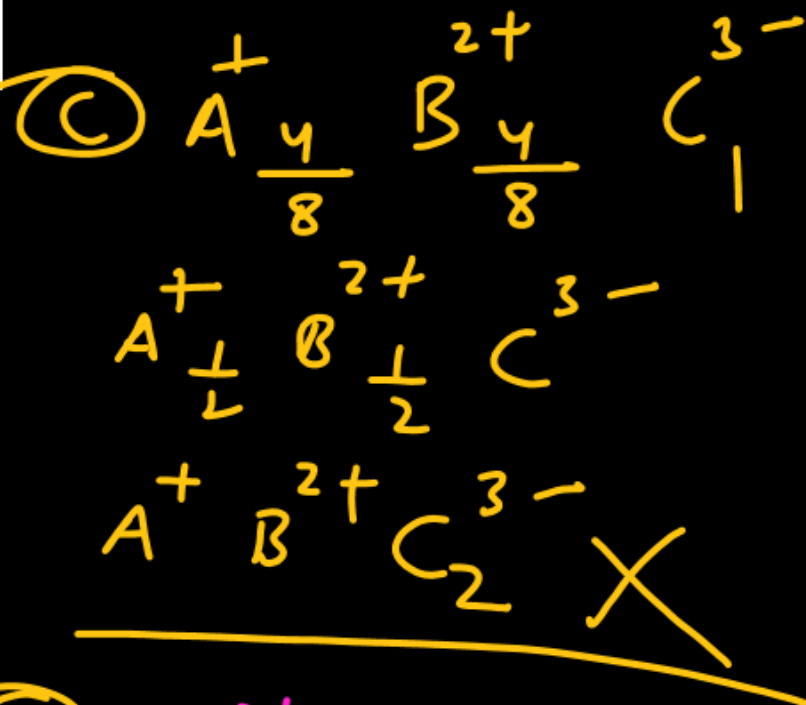
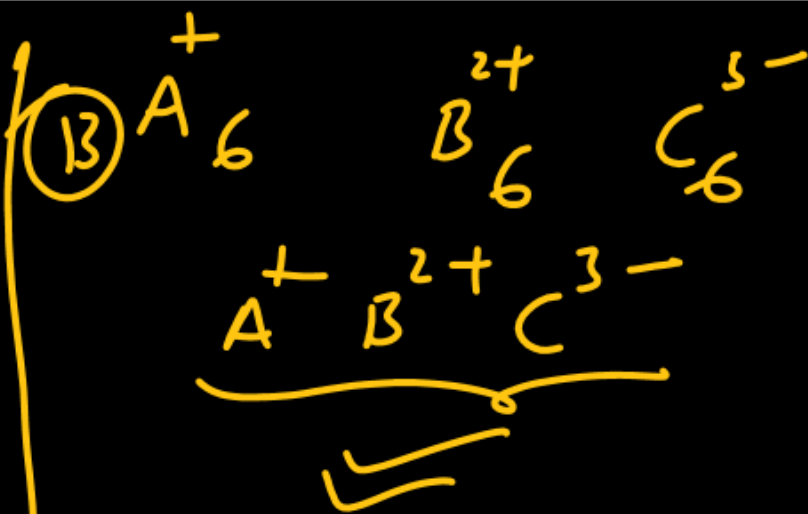
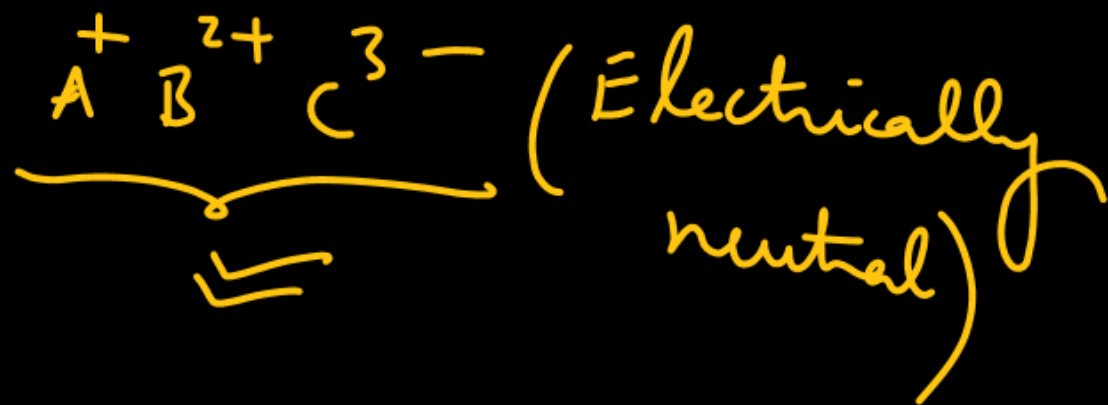
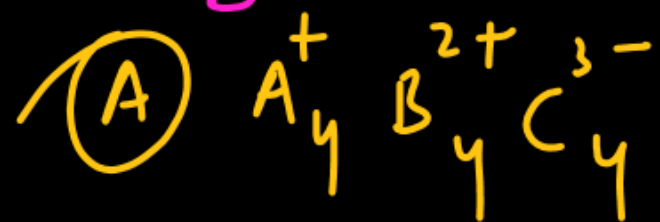
(C) $\rightarrow R$.
 Simple Cubic + body Centre
 (Cubical void)

$$P.f = \frac{1 \times \frac{4}{3} \pi R^3 + 1 \times \frac{4}{3} \pi (0.732R)^3}{\left(\frac{2R + 2 \times 0.732R}{\sqrt{3}} \right)^3}$$

$a\sqrt{3} = 2R + 2 \times 0.732R$
 $\rightarrow 0.732R$

An ionic compound consists of only A^+ , B^{2+} and C^{3-} ions. Which of the following option(s) is/are correct with respect to their possible structure ?

- ✓ (A) C^{3-} ions form CCP, A^+ ions occupy all octahedral voids and B^{2+} ions occupy half of the tetrahedral voids.
 ✓ (B) B^{2+} ions form HCP, A^+ ions occupy all octahedral voids and C^{3-} ions occupy half of the tetrahedral voids.
 ✗ (C) A^+ ions occupy alternate corners of a simple cube, B^{2+} ions occupy the other corners of the cube and C^{3-} ions occupy body centre of the cube.
 ✗ (D) A^{2+} ions form CCP, C^{3-} ions occupy all the tetrahedral voids and A^+ ions occupy all the octahedral voids.

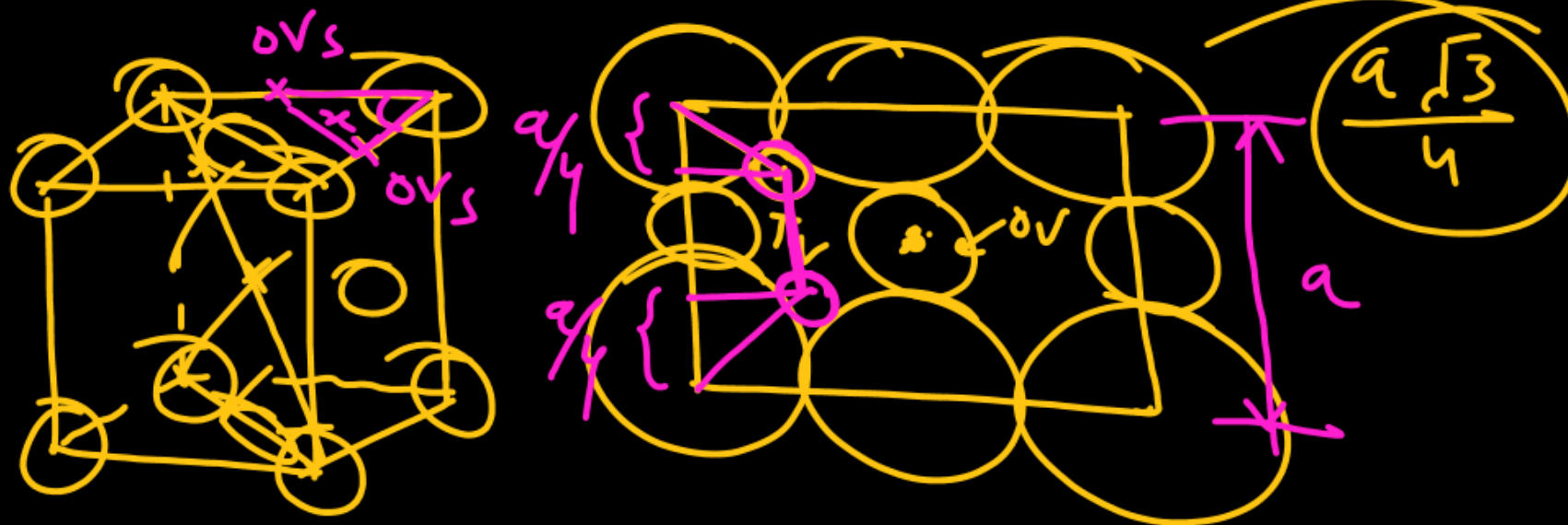


The correct information regarding a solid element (atomic radius = r) having ABCABC.... type packing (FCC unit cell edge length = a) is

- ✓ (A) the distance between nearest octahedral and tetrahedral void is $\frac{\sqrt{3}a}{4}$.
- ✓ (B) the distance between two nearest octahedral void is $\frac{a}{\sqrt{2}}$.
- ✗ (C) the distance between two nearest tetrahedral void is $\frac{\sqrt{3}a}{2}$.
- ✓ (D) the distance between successive A and B layers is $2r\sqrt{\frac{2}{3}}$.

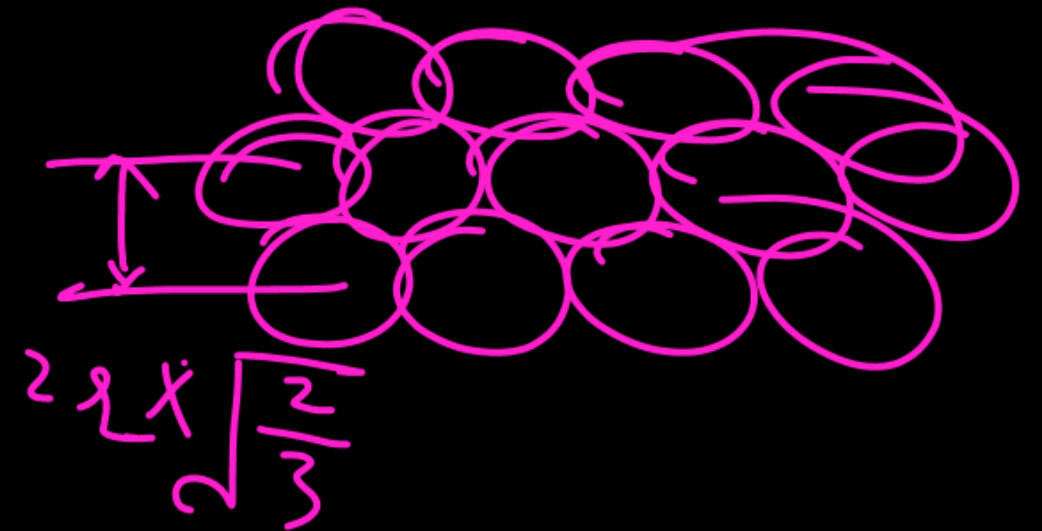
Nearest distance
between two TVs
 $= a/2$

Solⁿ: ABCABCABC... (CCP or FCC)



$$x^2 = \left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2$$

$$x = \frac{a}{\sqrt{2}}$$



An element exists in two allotropic forms. One form is CCP and the other form is HCP arrangement of atoms of the element. If the atomic radius of element is same in both the forms, then both allotropic forms have the same

(A) density (same)

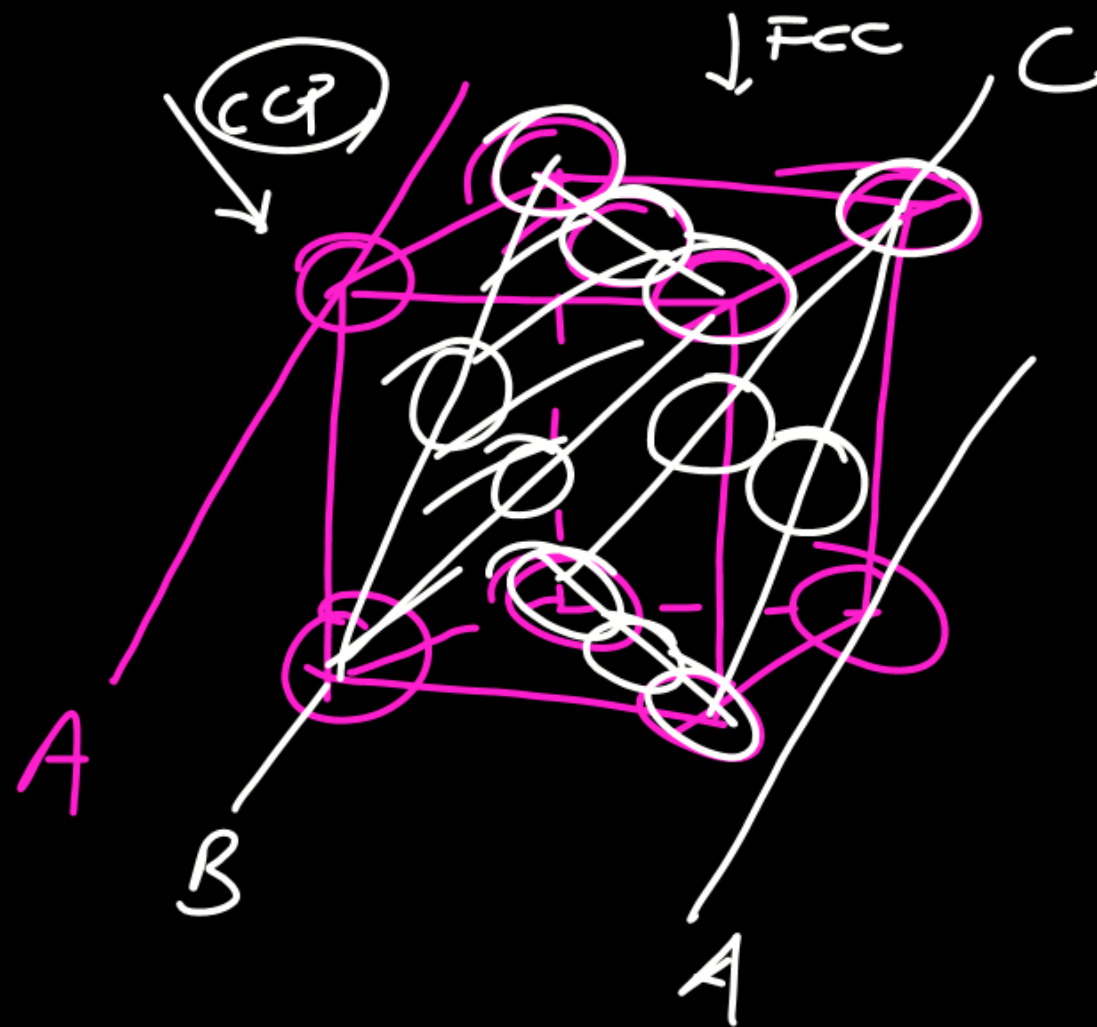
(C) fraction of unoccupied space.

(B) coordination number.

(D) distance between two consecutive layers.

$CN = 12$ in both.

$$(100 - 74)\% = 26\%$$



$$22 \times \sqrt{\frac{2}{3}}$$

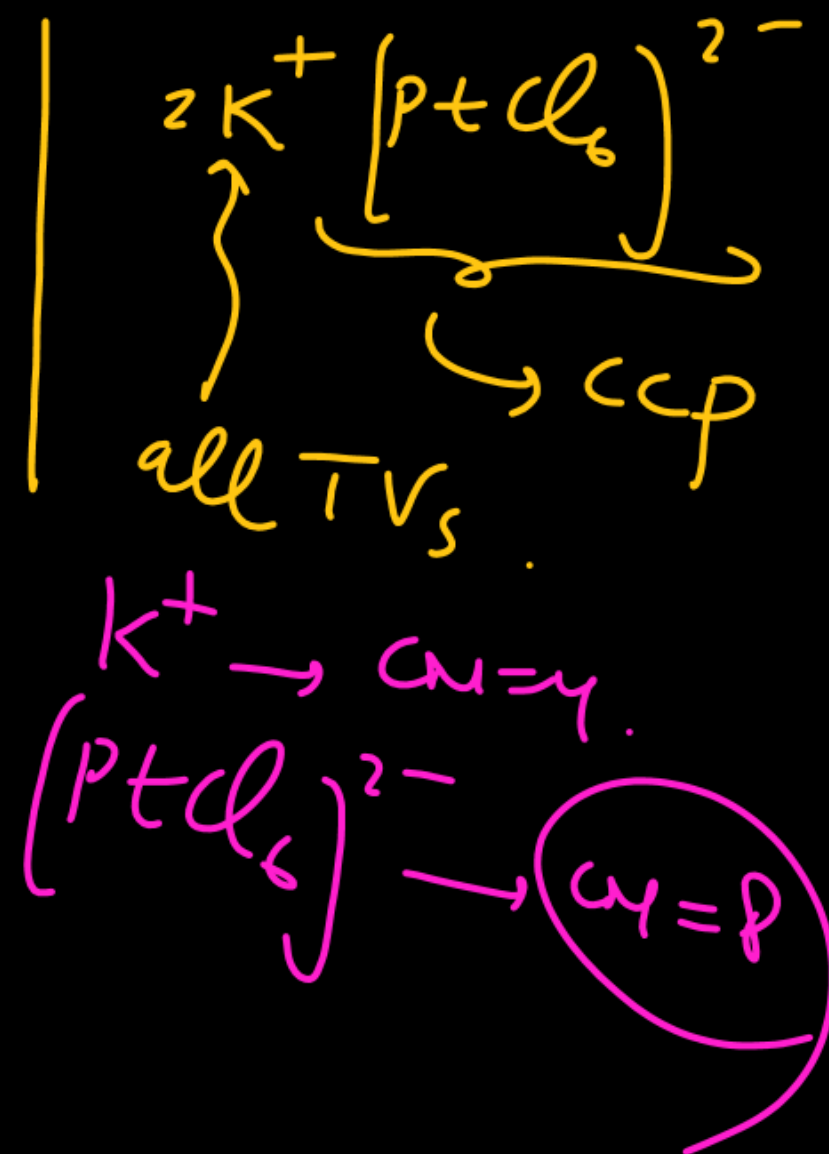
→ same in HCP and CCP.

Potassium hexachloroplatinate (IV), $K_2[PtCl_6]$ adopts anti-fluorite crystal structure. Which of the following information is correct about ideal $K_2[PtCl_6]$ crystal?

- ☒ (A) K^+ ions occupy all the tetrahedral voids.
- ☒ (B) The unit cell is FCC with respect to $PtCl_6^{2-}$ ions.
- ☒ (C) The coordination number of $PtCl_6^{2-}$ ions is 4.
- ☒ (D) The coordination number of $PtCl_6^{2-}$ ions is 8.

Ans : A, B, D

Solⁿ: Antifluorite (Na_2O) str.
anion : CCP
Cation : All TVs



Which of the following statement is correct regarding defects in solid?

- ☒ (A) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion.
- ☒ (B) Frenkel defect is a dislocation defect.
- ☒ (C) Trapping of an electron in the lattice leads to the formation of F-centre.
- ☒ (D) Schottky defect have no effect on the physical properties of solids.

Solⁿ:

Am: B & C

V.V. Imp

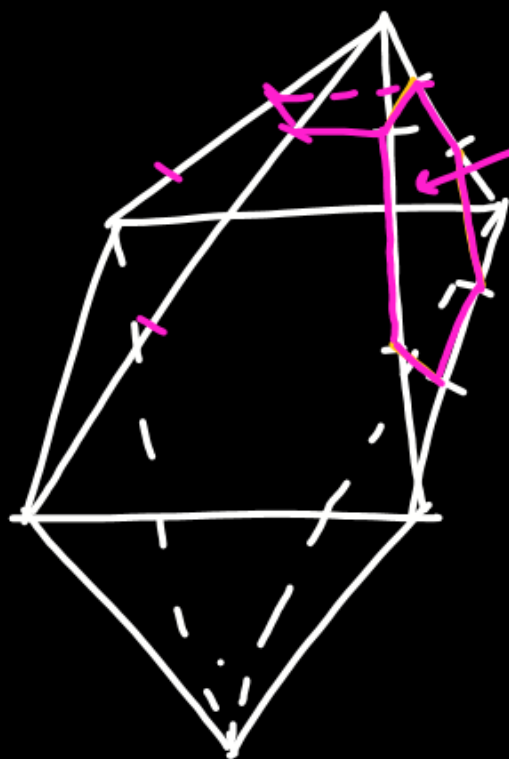
Number of square faces in truncated octahedron = x

Number of hexagonal faces in truncated tetrahedron = y

$x + y$ is

$$x + y = 6 + 4 = 10$$

JEK-Adv
(Similar)



hexagonal face.

No. of square faces (x) = 6.

No. of hexagonal faces = 8.

(JEK-Adv)

Octahedron
8 faces.



Truncated
Tetrahedron

hexagonal face.

Hexagonal faces: $y = 4$

Triangular faces: 4

